

Applying we get:

$$f(a) = \sum_{i=1}^n L(y_i, \hat{y}_i^{(t-1)})$$

gradient g_i

$$\frac{\partial}{\partial w}$$

$$f'(a) (n-a) = \sum_{i=1}^n \left[\frac{\partial L(y_i, \hat{y}_i^{(t-1)})}{\partial \hat{y}_i^{(t-1)}} \right] f_t'(x_i)$$

$$\frac{f''(a) (n-a)^2}{2} = \frac{1}{2} \sum_{i=1}^n \left[\frac{\partial^2 L(y_i, \hat{y}_i^{(t-1)})}{\partial \hat{y}_i^{(t-1)2}} \right] f_t''(x_i)$$

$\left(\frac{\partial^2}{\partial \hat{y}_i^2} \right)$ Hessian h_i .

$$\mathcal{L}^{(t)} = \sum_{i=1}^n \left[L(y_i, \hat{y}_i^{(t-1)}) + g_i f_t(x_i) + \frac{1}{2} h_i f_t^2(x_i) \right] + \Omega(f_t(x_i))$$

Here $L(y_i, \hat{y}_i^{(t-1)})$ is a constant.

Later, when we will do diff then it will be automatically zero later.

then expand $\Omega(f_t(x_i))$

$$\Omega(f) = Y^T + \frac{1}{2} \lambda \cdot \|w\|^2$$

↓

$$\frac{1}{2} \lambda (w_1^2 + w_2^2 + \dots)$$

$$\mathcal{L}^{(t)} = \underbrace{\sum_{i=1}^T}_{\# \text{ nodes}} \left(\underbrace{g_i}_{\omega_j} f_i(\mathbf{x}_i) + \frac{1}{2} h_i f_i^2(\mathbf{x}_i) \right) + VT + \frac{1}{2} \lambda \left[\underbrace{\sum_{j=1}^T \omega_j^2}_{\# \text{ leaf nodes}} \right]$$

$\omega_j = f_j(\mathbf{x}_i)$

$$= \sum_{i=1}^T \rightarrow \sum_{j=1}^T \sum_{i \in I_j}$$

Instance set of leaf j.
how much dataset fall in that leaf node.

$$\Rightarrow \mathcal{L}^{(t)} = \sum_{j=1}^T \left[\sum_{i \in I_j} g_i \omega_j + \frac{1}{2} \sum_{i \in I_j} h_i \omega_j^2 \right] + VT + \frac{1}{2} \lambda \sum_{j=1}^T \omega_j^2$$

$$\Rightarrow \mathcal{L}^{(t)} = \sum_{j=1}^T \left[\sum_{i \in I_j} g_i \omega_j + \sum_{i \in I_j} h_i \omega_j^2 + \frac{1}{2} \lambda \omega_j^2 \right] + VT$$

$$\Rightarrow \mathcal{L}^{(t)} = \sum_{j=1}^T \left[\sum_{i \in I_j} g_i \omega_j + \frac{1}{2} \left(\sum_{i \in I_j} h_i + \lambda \right) \omega_j^2 \right] + VT$$

$$\Rightarrow \frac{\partial \mathcal{L}^{(t)}}{\partial \omega_j} = \sum_{i \in I_j} g_i + \left(\sum_{i \in I_j} h_i + \lambda \right) \omega_j = 0$$

$$\omega_j = \frac{- \sum_{i \in I_j} g_i}{\sum_{i \in I_j} h_i + \lambda} \rightarrow \text{gradient.}$$

$\hookrightarrow$ Hessian.

Regression: $\rightarrow \text{mse} = L = \frac{1}{2} \sum_{i=1}^n (y_i - \hat{y}_i)^2$

$$g_i = \frac{\partial L}{\partial \hat{y}_i} = \sum_{i=1}^n (\hat{y}_i - y_i)$$

$$h_i = 1$$

$$w_j = \frac{\sum_{i \in I_j} (y_i - \hat{y}_i)}{\sum_{i \in I_j} 1 + L} \rightarrow \begin{array}{l} \text{sum of residuals} \\ \text{\#no. of residuals} + L \end{array}$$

$$\therefore w_j = \frac{\text{sum of residuals}}{\text{\#no. of residuals} + L}$$

Classification $\rightarrow \log(\text{odd})$

$$g_i = (y_i - p_i) \rightarrow \text{probability (residual)}$$

$$h_i = p_i(1 - p_i)$$

$$w_j = \frac{\text{sum of residuals}}{\sum_{i \in I_j} p_i(1 - p_i) + L}$$